

KWAJALEIN ATOLL, Marshall IslandsWMO: **913660**Lat: **8.720N**Lon: **167.730E**Elev: **2**StdP: **101.30**Time Zone: **12.00 (E12)**Period: **94-19**WBAN: **40604****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	24.6	25.0	18.9	13.7	27.9	19.8	14.5	28.0	12.1	27.2	11.4	27.4	5.5	70	0.612

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
9	4.1	31.4	26.0	31.2	25.9	30.9	25.8	27.5	29.9	27.2	29.7	26.9	29.5	5.2	70

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
26.8	22.4	29.1	26.4	21.8	29.0	26.1	21.5	28.9	87.3	30.0	85.7	30.0	84.6	29.8	29.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.2	10.4	9.8	23.5	31.8	0.5	0.5	23.2	32.2	22.9	32.4	22.6	32.7	22.3	33.0		
(5)			20.8	27.9	0.8	1.0	20.2	28.6	19.7	29.2	19.2	29.8	18.6	30.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	28.2	27.9	27.9	28.1	28.3	28.3	28.2	28.2	28.4	28.3	28.2	28.2	
	DBStd	0.80	0.66	0.63	0.70	0.76	0.79	0.81	0.84	0.85	0.87	0.88	0.87	
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	
	CDD10.0	6634	554	500	562	548	568	548	563	565	552	566	546	
	CDD18.3	3592	296	267	303	298	310	298	305	307	302	307	296	
	CDH23.3	41464	3233	2887	3388	3426	3619	3494	3576	3621	3618	3652	3463	
	CDH26.7	13017	828	726	969	1076	1187	1145	1167	1212	1280	1242	1129	
Wind	WSAvg	6.0	7.5	7.5	7.2	7.1	6.4	5.9	5.0	4.1	4.0	4.3	5.6	
Precipitation	PrecAvg	2547	115	90	114	190	225	226	257	269	279	302	273	
	PrecMax	3720	367	351	594	515	682	400	566	598	560	509	488	
	PrecMin	1507	12	1	4	5	13	52	90	137	96	106	89	
	PrecStd	531	93	77	115	132	144	86	80	106	97	101	101	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	30.7	30.7	31.1	31.2	31.2	31.2	31.4	31.6	31.7	31.7	31.5	
		MCWB	25.0	24.9	25.2	25.1	25.6	25.9	26.0	26.1	26.2	26.3	26.3	
	2%	DB	30.2	30.2	30.7	30.9	31.0	31.0	31.0	31.1	31.2	31.1	31.0	
		MCWB	24.9	24.7	25.0	25.1	25.6	25.8	26.0	26.0	26.2	26.2	26.1	
	5%	DB	29.9	29.9	30.2	30.4	30.5	30.5	30.5	30.7	30.9	30.7	30.5	
		MCWB	24.8	24.6	25.0	25.2	25.6	25.8	26.0	26.0	26.1	26.1	26.0	
	10%	DB	29.4	29.5	29.9	30.0	30.1	30.1	30.1	30.2	30.4	30.2	30.1	
		MCWB	24.8	24.6	24.9	25.2	25.7	25.8	26.0	26.0	26.0	26.0	25.9	
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	27.0	26.9	26.9	27.1	27.3	27.3	27.5	27.6	28.4	27.7	27.6	
		MCDB	29.4	29.4	29.9	29.8	29.9	29.7	29.8	30.2	30.5	30.0	29.8	
	2%	WB	26.4	26.2	26.2	26.5	26.9	27.0	27.0	27.1	27.2	27.1	27.1	
		MCDB	28.9	28.8	29.2	29.4	29.5	29.6	29.6	29.8	29.8	29.7	29.6	
	5%	WB	26.0	25.7	25.9	26.2	26.6	26.6	26.7	26.7	26.7	26.7	26.7	
		MCDB	28.5	28.5	28.9	29.1	29.2	29.3	29.3	29.5	29.5	29.5	29.3	
	10%	WB	25.6	25.4	25.6	25.9	26.2	26.3	26.4	26.4	26.5	26.5	26.4	
		MCDB	28.2	28.2	28.6	28.8	28.9	29.0	29.1	29.2	29.3	29.2	29.0	
Mean Daily Temperature Range	5% DB	MDBR	3.3	3.5	3.6	3.6	3.6	3.7	3.8	4.0	4.1	4.0	3.7	
		MCDBR	3.5	3.7	3.9	3.8	3.8	3.9	4.1	4.3	4.3	4.3	4.1	
		MCWBR	2.0	2.0	1.9	1.9	1.8	1.8	2.0	2.0	2.1	2.1	2.1	
	5% WB	MCDBR	3.4	3.6	3.7	3.7	3.7	3.8	3.9	4.1	4.2	4.1	3.9	
		MCWBR	2.3	2.2	2.1	2.2	2.1	2.0	2.3	2.3	2.4	2.3	2.3	
Clear-Sky Solar Irradiance	taub	0.439	0.443	0.445	0.456	0.437	0.428	0.415	0.394	0.394	0.399	0.437	0.457	
	taud	2.385	2.374	2.395	2.383	2.467	2.499	2.548	2.617	2.617	2.590	2.460	2.329	
	Ebn at Noon	873	881	879	858	857	855	869	900	913	911	870	847	
	Edh at Noon	122	126	125	125	112	107	103	97	99	100	112	126	
All-Sky Solar Radiation	RadAvg	5.33	5.83	6.14	6.08	5.83	5.70	5.59	5.74	5.63	5.24	4.90	5.03	
	RadStd	0.20	0.24	0.41	0.43	0.42	0.29	0.28	0.47	0.30	0.36	0.32	0.24	